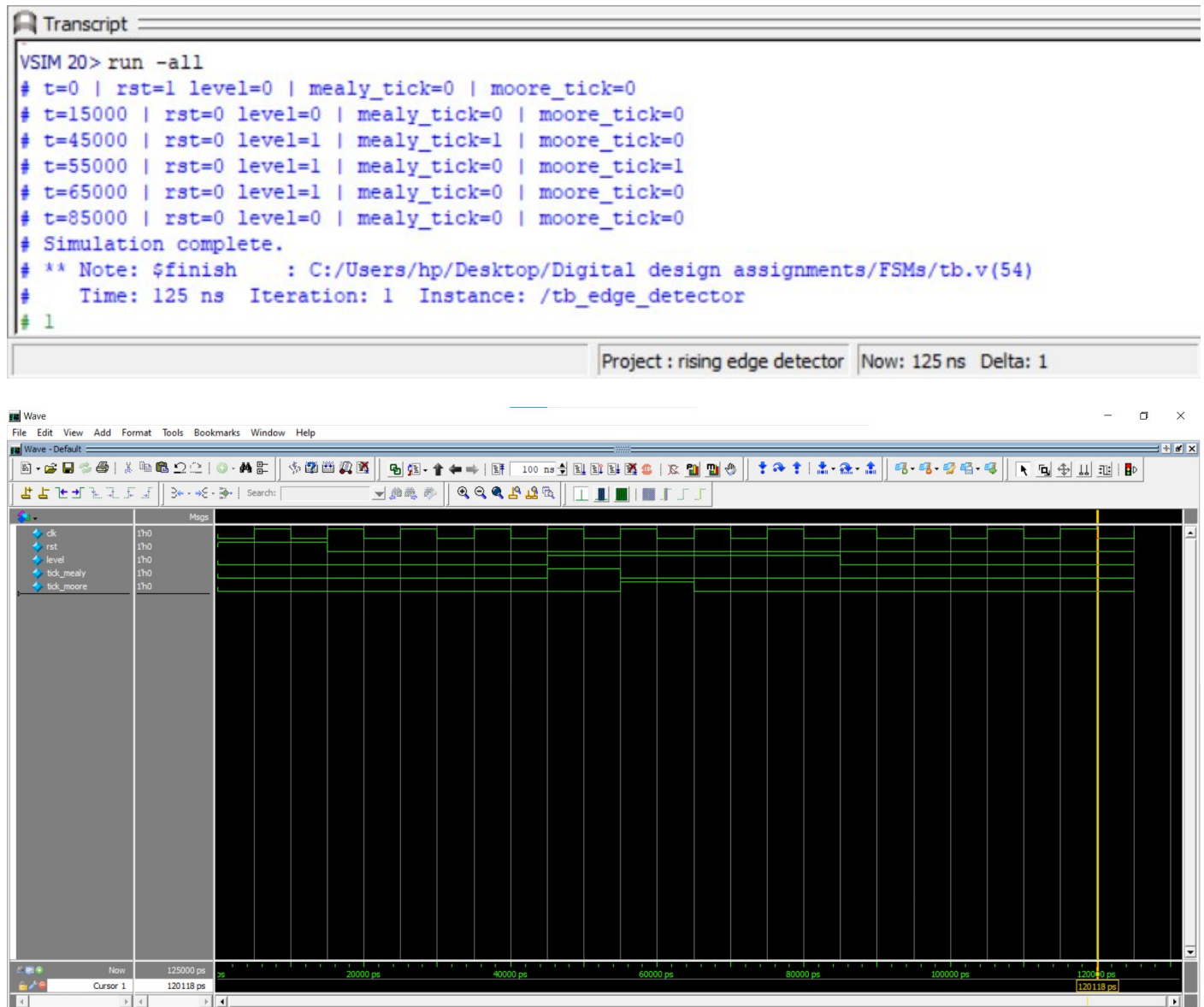


Rising edge detector

Result:

All test cases produced outputs matching the expected values:



The Mealy machine detects the rising edge combinationaly, so tick pulses in the exact same instant level transitions high ($t=45000\text{ns}$) — no delay, since its output depends on both state and input.

The Moore machine, whose output depends on state alone, requires the extra transient EDG state to represent "edge just occurred", so its tick pulses one full clock cycle later ($t=55000\text{ns}$). Both pulses are exactly one clock period wide and return to 0 correctly once the FSM advances past the triggering condition, and both stay idle correctly during reset and when level is steady.

This matches the timing diagram from the original design spec precisely, confirming the classic trade-off: **Mealy is faster (zero-cycle latency) but Moore is glitch-free and purely state-driven**, at the cost of one cycle of latency and an extra state.